MAJOR PROJECT REPORT

AI-BASED NETWORK INTRUSION DETECTION USING UNSUPERVISED MACHINE LEARNING FOR ANOMALY DETECTION

ABSTRACT

This project focuses on detecting abnormal network activities using machine learning techniques. The NSL-KDD dataset is used to analyze network traffic and identify suspicious behavior. Data preprocessing methods such as label encoding and feature scaling are applied before training the model. The Isolation Forest algorithm is used to detect anomalies without requiring labeled training data. The system helps improve cybersecurity by identifying potential intrusions and unusual network activities.

INTRODUCTION

Network security is an important aspect of modern computing systems. Cyberattacks such as unauthorized access, malware attacks, and data breaches are increasing rapidly. Traditional intrusion detection systems rely on predefined attack signatures and may fail to detect unknown threats.

This project uses the Isolation Forest algorithm, an unsupervised machine learning technique, to detect anomalies in network traffic. The model analyzes network behavior and identifies suspicious activities that may indicate cyberattacks.

OBJECTIVE

- Detect abnormal network activities.
- Analyze network traffic using machine learning.
- Apply Isolation Forest for anomaly detection.
- Improve network security through intelligent monitoring.

TOOLS USED

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Scikit-Learn
- GitHub

METHODOLOGY

Data Loading

The NSL-KDD dataset is loaded into Jupyter Notebook using Pandas.

Data Preprocessing

The dataset is checked and prepared for machine learning analysis.

Label Encoding

Categorical values such as protocol type and service are converted into numerical form.

Feature Scaling

StandardScaler is used to normalize the dataset.

Isolation Forest

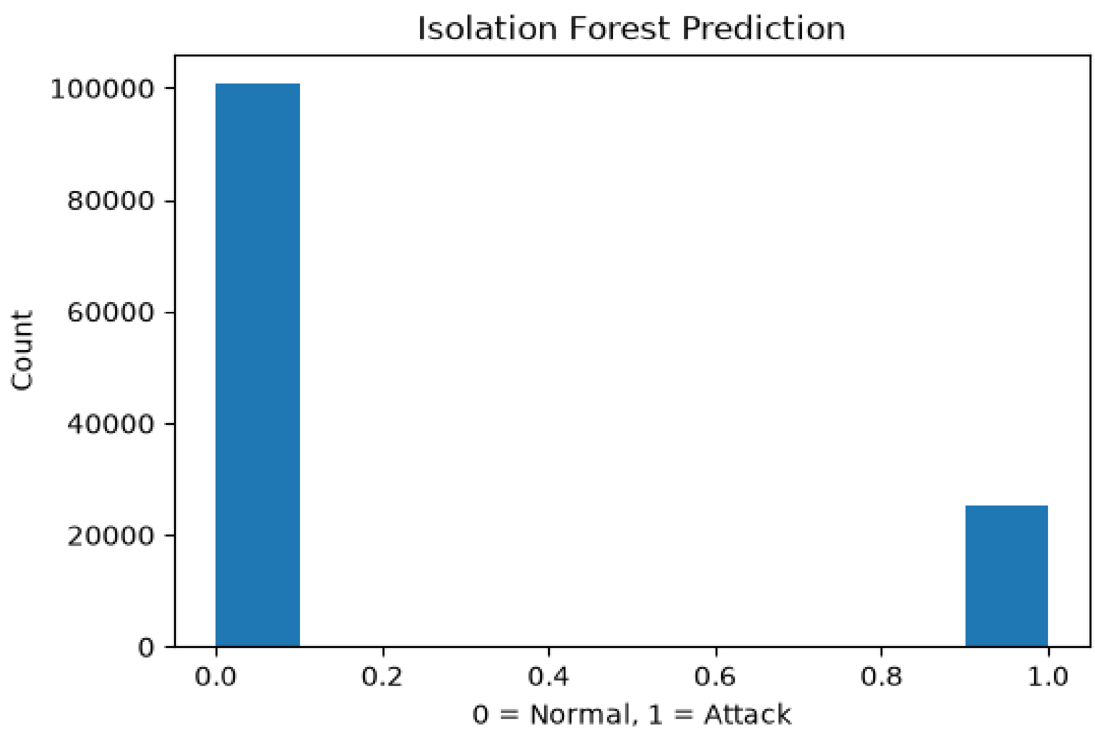
The Isolation Forest algorithm is applied to detect anomalies in network traffic.

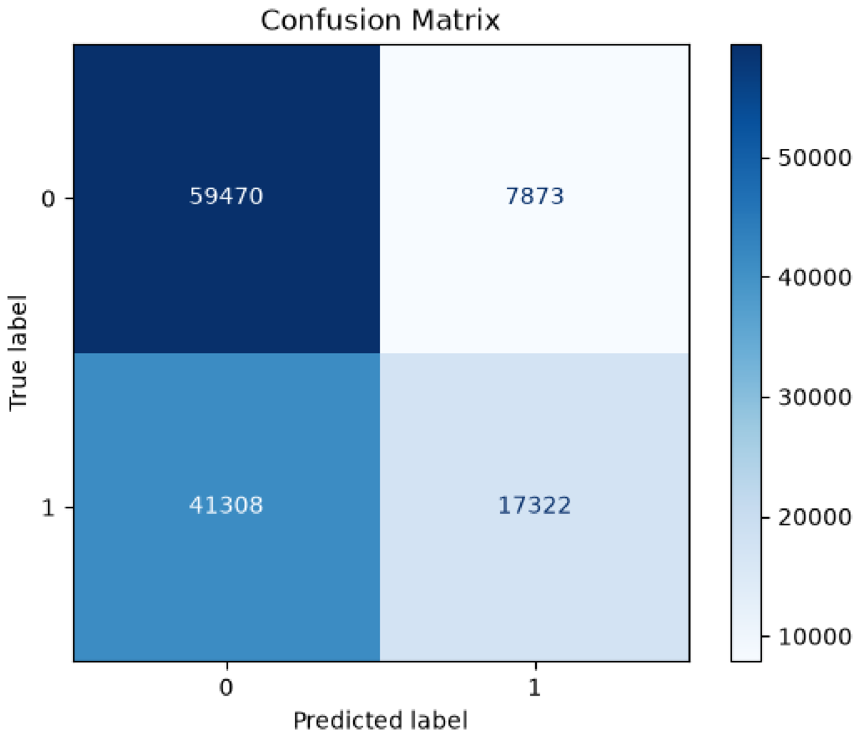
Result Analysis

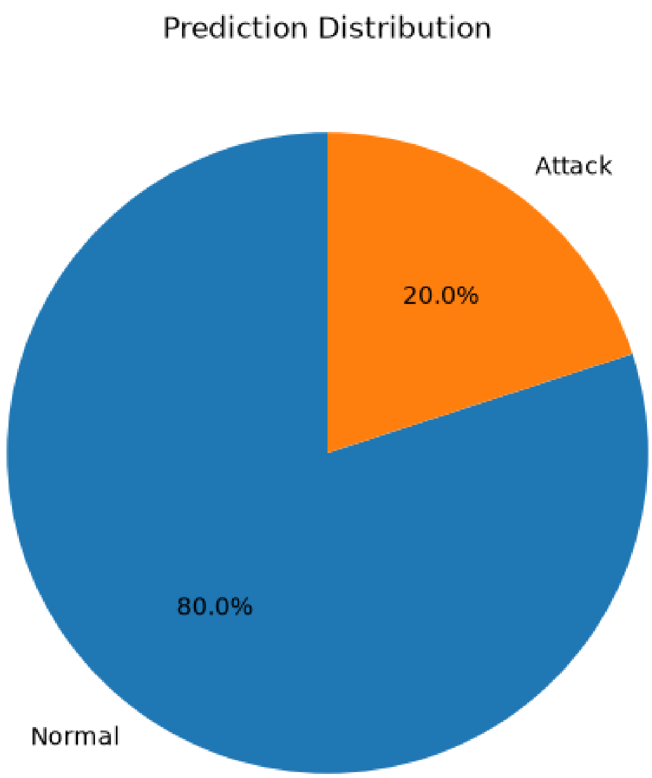
The detected anomalies are analyzed and visualized

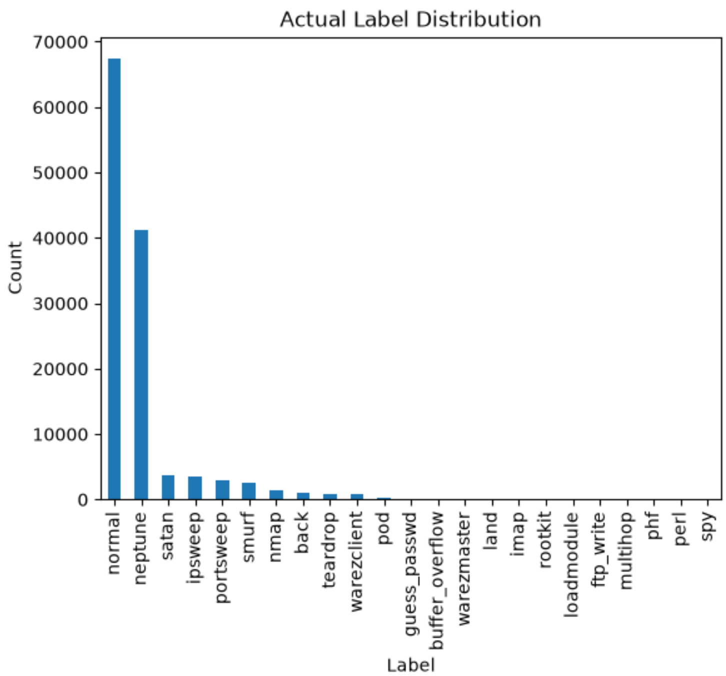
RESULTS

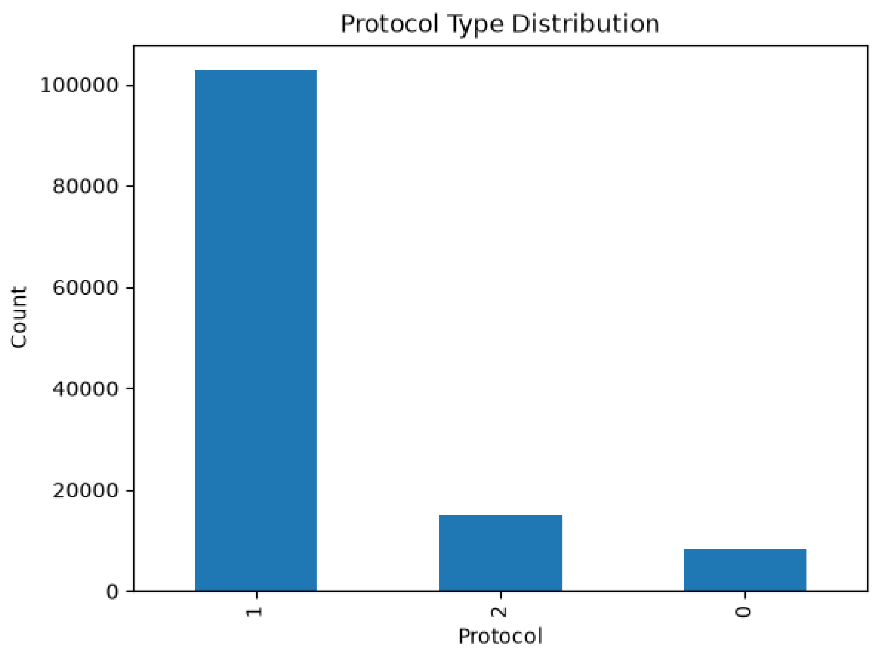
The NSL-KDD dataset was successfully processed and analyzed using the Isolation Forest algorithm. After data preprocessing, label encoding, and feature scaling, the model was able to identify anomalous network activities. The results showed that the algorithm effectively distinguished normal traffic from suspicious behavior, demonstrating the usefulness of unsupervised machine learning in network intrusion detection.

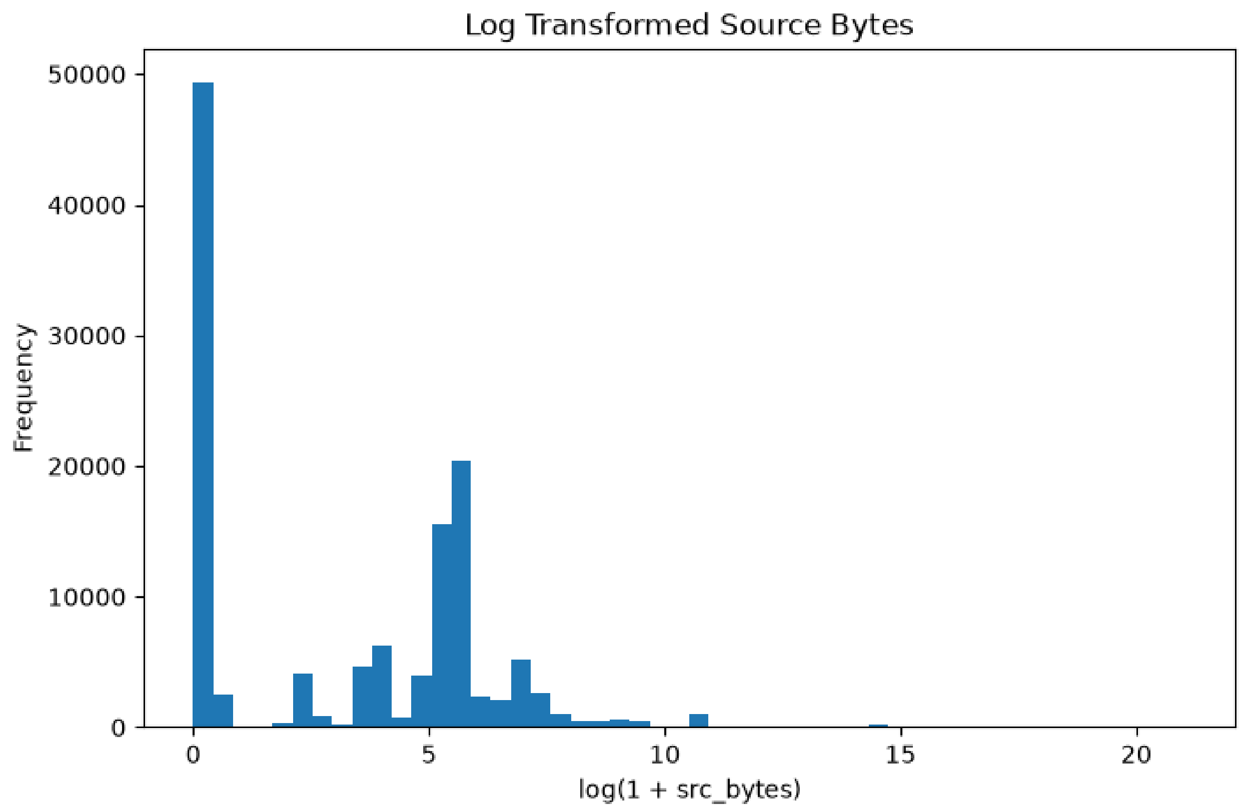



Confusion Matrix
0
1
True label
59470
7873
41308
17322
0
1
Predicted label
50000
40000
30000
20000
10000

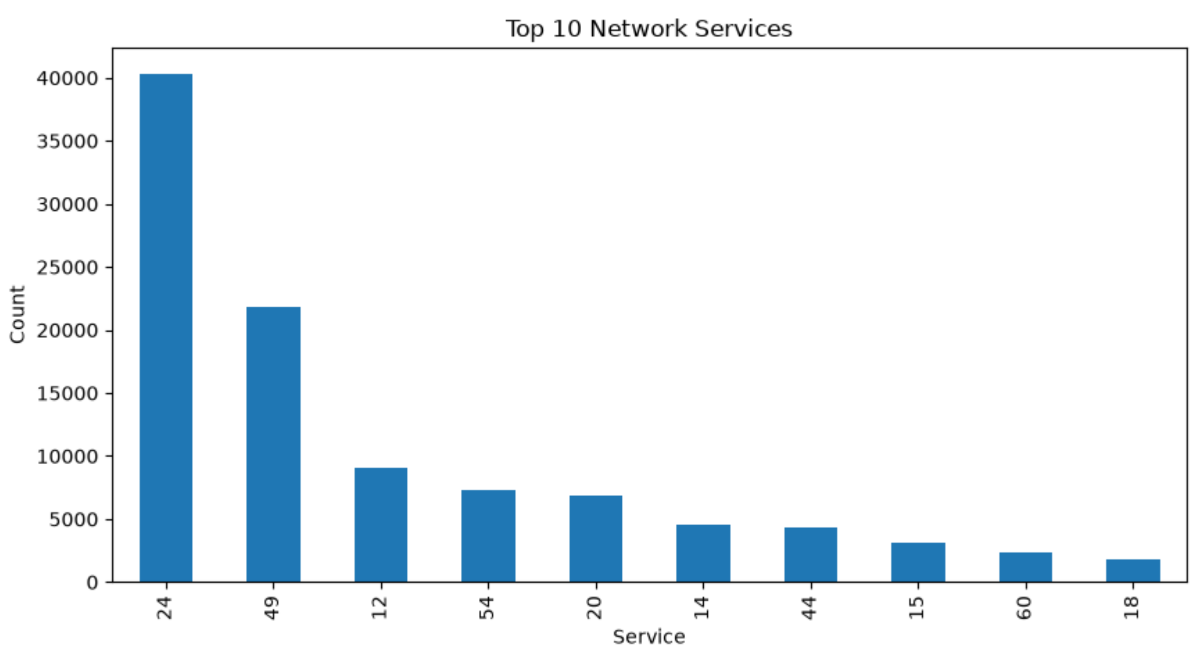

Prediction Distribution
Attack
20.0%
80.0%
Normal


Actual Label Distribution
70000
60000
50000
40000
30000
20000
10000
0
Count
normal
neptune
satan
ipsweep
portsweep
smurf
nmap
back
teardrop
warezclient
pod
guess_passwd
buffer_overflow
warezmaster
land
imap
rootkit
loadmodule
ftp_write
multihop
phf
perl
spy
Label


Protocol Type Distribution
100000
80000
60000
40000
20000
0
Count
1
2
0
Protocol



Log Transformed Source Bytes
Frequency
50000
40000
30000
20000
10000
0
0
5
10
15
20
log(1 + src_bytes)



Top 10 Network Services
Count
40000
35000
30000
25000
20000
15000
10000
5000
0
24
49
12
54
20
14
44
15
60
18
Service

Conclusion

This project demonstrates the application of artificial intelligence in network security. The Isolation Forest algorithm successfully detected anomalies in network traffic without requiring labeled training data. The proposed system can help improve cybersecurity by identifying suspicious activities and potential intrusions.

References

1. NSL-KDD Dataset
2. Python Documentation
3. Scikit-Learn Documentation
4. Jupyter Notebook Documentation